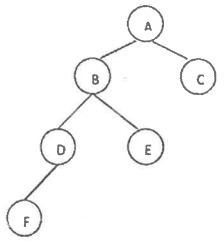
USN | | | | | | | | | | [23CSE105]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

DATA STRUCTURES USING C

Time: 03 HOURS **Maximum Marks:100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Answer all questions.

		PART - A			**05 X 02=10**
1.	**a.**	Is bound checking is compulsory in array?	L2	CO1	**02**
	b.	What is primitive data structures in c?	L1	CO2	**02**
	c.	What is the postfix form of the expression (A+ B)*(C*D– E)*F / G?	L3	CO3	**02**
	d.	Write any four applications of Queue.	L2	CO4	**02**
	e.	Define length of the path in a tree with an example.	L2	CO5	**02**
		PART – B			**06 X 05=30**
2.	**a.**	What is pointer? Elucidate pointer initialization with example.	L2	CO1	**05**
		OR			
	b.	Explain Storage classes in C language.	L1	CO1	**05**
3.	**a.**	Elucidate the types of data structures in C language.	L1	CO2	**05**
		OR			
	b.	What are the four dynamic memory allocation functions? Explain.	L2	CO2	**05**
4.	**a.**	Explain push and pop operation on stack.	L2	CO3	**05**
		OR			
	b.	Convert the given infix expression 2*3/(2-1)+5*(4-1) into postfix expression.	L3	CO3	**05**
5.	**a.**	What is queue? Explain representation of queue.	L2	CO4	**05**
		OR			
	b.	Explain the operations performed on queue data structure.	L2	CO4	**05**
6.	**a.**	Perform Preorder, Inorder, Postorder Traversal of the given Tree 	L3	CO5	**05**
		OR			
	b.	List the applications of tree data structure.	L2	CO5	**05**
7.	**a.**	Examine about the operations of Binary Search Tree.	L3	CO5	**05**

		OR			
	b.	Differentiate between array and linked list in data structure.	L2	CO2	**05**
		PART-C		**06 × 10=60**	
8.	**a.**	Describe array in C language.	L2	CO1	**10**
		OR			
	b.	Write a program for passing argument by value and passing argument by reference.	L3	CO1	**10**
9.	**a.**	Discuss in detail about insert and delete operation at any position for singly linked list with example.	L3	CO2	**10**
		OR			
	b.	Examine Doubly Linked List Operation.	L3	CO2	**10**
10.	**a.**	Recall the implementation of stack using array.	L2	CO3	**10**
		OR			
	b.	List the Types of graph and explain any 4 types.	L2	CO3	**10**
11.	**a.**	Explain the implementation of queue using linked list.	L3	CO4	**10**
		OR			
	b.	Illustrate types of queue.	L3	CO4	**10**
12.	**a.**	Develop the binary search tree for the following inputs 4,15,9,7,18,3,5,16,20,17,14.	L3	CO5	**10**
		OR			
	b.	Mention and explain the ways of representing a graph.	L2	CO5	**10**
13.	**a.**	Outline the steps for evaluation of postfix expression using stack with an example. Evaluate the following postfix expression: 5 9 3 + 4 2 * * 7 + *.	L3	CO3	**10**
		OR			
	b.	Organize the steps in constructing a minimal spanning tree using kruskal's algorithm and apply the algorithm for weighted undirected graph.	L3	CO5	**10**